ECO-METROPOLIS

Building a Sustainable Tomorrow, One Tile at a Time

Interactive Educational Board Game for Environmental Preservation

End Semester Project Report

Course: ES204L Ecology, Environment and Sustainability

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Abstract

This project introduces an interactive, hybrid physical-digital board game designed to simulate the intricate balance between state governance and environmental preservation. The simulation places participants in the role of Mayors managing underdeveloped cities. The primary objective is to reach a 4,000-point threshold to transition into "developed" status, and ultimately secure victory by surviving the advanced inner ring of the board. Unlike traditional tabletop games focused solely on wealth accumulation, victory in this simulation requires the strategic management of four weighted key metrics: Smartness, Livability, Sustainability, and Economy. Players select from four distinct state archetypes—Natural, Industrial, Tech, or Financial—each offering unique resource advantages and metric weightings. The gameplay mechanics feature infrastructure tiles and event cards that force players to make critical socio-economic trade-offs. To enhance immersion and eliminate manual calculations, the game utilizes a sophisticated IoT framework. This includes an ESP32 microcontroller, magnetic encoders to track physical movement, RFID readers for card interactions, and a mobile application integrated via WebSockets. This digital-physical hybrid ensures a seamless educational experience that elucidates the real-world consequences of industrial, social, and ecological policies.

1 Introduction and Literature Review

The challenge of educating younger audiences—and the general public—about the inner workings of government, urban planning, and environmental stewardship remains significant. Complex systems involving supply chains, carbon emissions, and public infrastructure are often too abstract to grasp through traditional educational media. This project introduces a "serious game" simulation designed to clarify state management by placing players directly in decision-making roles. By casting players as state representatives, the game fosters an immediate awareness of governmental functions while emphasizing the critical, often conflicting mandate to protect the environment while driving economic growth. The simulation creates an instructive atmosphere wherein players must navigate a dynamic environment, utilizing both competition and cooperation with other Mayors to build an "ideal world" where development thrives alongside ecological preservation.

A review of existing tabletop gaming literature and market offerings reveals the foundational mechanics that inspired this project, alongside a distinct gap in current designs. The development of this project is deeply rooted in the mechanics of traditional roll-and-move board games, specifically drawing inspiration from classics such as

Monopoly and *The Game of Life*. From *Monopoly*, this project adopts the core concepts of navigating a looped board, managing real estate (infrastructure tiles), and progressing through tiered stages of development.

However, traditional economic board games inherently postulate that financial accumulation is the sole metric for success. In *Monopoly*, the player with the most capital and property wins, regardless of the simulated societal cost. A review of existing tabletop games reveals a significant lack of comprehensive, nature-oriented simulations that address the multi-dimensional responsibilities of a modern state leader. This project differentiates itself by moving decisively beyond capital. While budget tokens are necessary for gameplay, the simulation demands the careful balancing of four distinct pillars: Smartness, Livability, Sustainability, and Economy.

Furthermore, modern board game design is increasingly exploring the integration of digital companion applications to handle complex state management. Games such as *Mansions of Madness* or *Descent* utilize applications to drive narrative and enemy artificial intelligence. This project expands upon that boundary by utilizing a localized Internet of Things (IoT) network. By offloading the complex, multi-variable calculus of urban planning to an ESP32 microcontroller and a companion Flutter application, the game evolves the traditional board game into a highly advanced, frictionless tool for civic and ecological education, without encumbering the players with manual bookkeeping.

2 Methodology

The methodology for developing this interactive simulation involves a client-server architecture bridging physical hardware and mobile software to create a localized, real-time gaming ecosystem.

System Architecture

The core logic engine of the game is powered by an ESP32 microcontroller acting as the localized server. The ESP32 hosts a WebSocket server to ensure real-time, low-latency communication between the physical board and the players' digital interfaces.

The physical board is equipped with several hardware inputs:

- **Movement Tracking:** A central 1-10 spinner is attached to a magnetic encoder. When a player spins the wheel, the encoder transmits the raw data to the ESP32, which calculates the player's new position on a one-dimensional mathematical track representing the board's spiral grid.

- **Card Interactions:** An RFID reader is integrated into the board. This is utilized to scan physical Special Cards (Policies, Events, Disasters) equipped with RFID tags.

Four players connect to the ESP32 via a Flutter-based mobile application. The application receives real-time JSON state updates and transmits player decisions back to the server. The application serves as the "Mayor's Terminal," replacing physical paper sheets by displaying real-time bank balances, dynamic radar charts for city factors, and interactive dialog modals for infrastructure purchases.

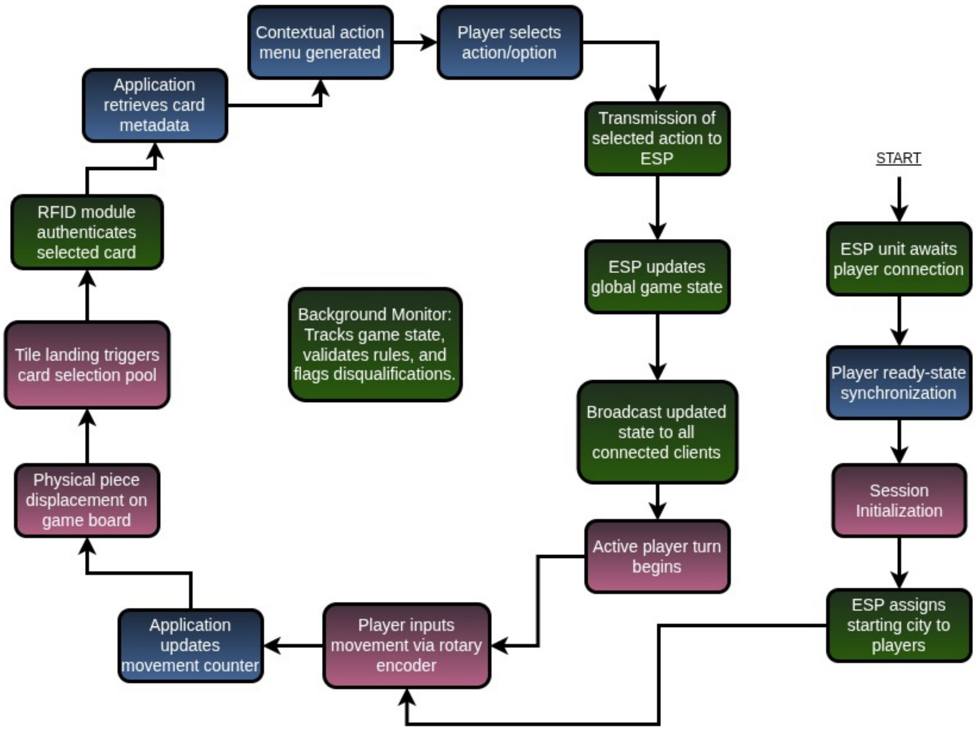

Figure 1: Control Flowchart

Game Mechanics and Mathematical Engine

Players are randomly assigned one of four City Hubs: Natural Resources, Software (Tech), Industrial, or Financial/Cultural. To necessitate asymmetric strategies, each city calculates its Total Score (S_{Total}) using uniquely weighted percentages (W_i) of the four main factors (F_i). For example, the Natural Hub weights Sustainability at 40%, whereas the Industrial Hub weights Economy at 50%.

The total score is expressed as the weighted sum of the four factors:

$$S_{Total} = \sum_{i=1}^{4} (W_i \times F_i)$$

The ESP32 maintains a digital ledger for each player, recalculating scores from scratch at the conclusion of every turn. The tiles on the board can represent infrastructure, a policy, an event, or a disaster. Each infrastructure purchase, policy implementation, or disaster occurrence directly alters the underlying metrics.

The system tracks 8 core metrics: Emission, Pollution Index, Happiness Index, Biodiversity Health, Community Trust, Tech Integration, Economic Output, and Infrastructure Efficiency. To ensure a positive correlation for the final factor scores, negatively impacting metrics (Emission and Pollution) are inverted using a predefined maximum scale value (M_{Max}). The four principal factors are mathematically derived as follows:

$$F_{Sustainability} = \frac{M_{Biodiversity} + (M_{Max} - M_{Emission}) + (M_{Max} - M_{Pollution})}{3}$$

$$F_{Smart} = \frac{M_{Tech} + M_{Infra}}{2}$$

$$F_{Livability} = \frac{M_{Happiness} + M_{Trust} + M_{Infra} + (M_{Max} - M_{Pollution})}{4}$$

$$F_{Economy} = \frac{M_{Output} + M_{Tech} + M_{Trust}}{3}$$

These metrics, upon reaching a certain threshold, can reward or penalize the player, rendering it imperative for a Mayor to continuously monitor their local indices. Players move across the board, landing on empty infrastructure tiles (which they may purchase to become service providers) or occupied tiles (where they must pay a service fee to the owner). Completing a lap deposits the player's current Economic Output metric into their Bank Balance as spendable Budget Tokens.

3 Results and Discussion

The implementation of the hybrid IoT framework results in a highly dynamic gameplay loop that successfully simulates real-world urban management without encumbering players with manual mathematical calculations.

Economic System and Infrastructure Dynamics

During simulated gameplay, the decision to purchase infrastructure proved to be a critical strategic pivot. There are 48 available tiles, ranging from Wind Farms to Data Centers, alongside special tiles for policies, events, or disasters. When a player lands on a tile and purchases it via their Flutter application, the ESP32 instantly applies immediate factor points and registers future base metric upgrades that trigger after specific lap intervals. The implementation of the "Service Provider" mechanic—where landing on an occupied tile forces a player to pay the owner—successfully mimics a localized economy and encourages players to monopolize specific sectors (such as Green Energy or Tech) to generate passive income.

Figure 2: Main Board

The Impact of Faction Asymmetry

The weighted scoring system profoundly impacts player behavior. In a simulated run, Player 1 (Software Hub) and Player 2 (Cultural/Tourism Hub) approached the board with divergent strategies. Because the Software Hub weights the "Smart" factor at 40%, Player 1 aggressively pursued Data Centers and Autonomous Vehicle Hubs. While this bolstered their Smart score, it caused their Sustainability (weighted at 10%) to decline. However, as long as no core metric fell below the critical baseline (which triggers elimination), the asymmetric weights allowed Player 1 to thrive. Conversely, Player 2 was forced to prioritize Livability and Sustainability, relying on Eco-Tourism and mass transit to maintain competitive equilibrium.

Special Tiles and Disaster Mechanics

The integration of RFID cards for Special tiles yielded the most significant narrative and educational impact. When players land on a Special tile, they draw and scan a physical card:

- **Policy Cards** (e.g., Carbon Tax) require players to make a strategic choice regarding policy implementation. The chosen parameters are instantly updated within the ESP32 ledger.
- **Event Cards** (e.g., Community Recycling) dictate forced events where the player must accept the consequences, or occasionally provide a secondary choice.
- **Disaster Cards** (e.g., Floods, Heatwaves) inflict immediate environmental penalties based dynamically on the city's current sustainability levels.

Furthermore, the system successfully executed "Cascading Penalties." Because cities operate as part of a global supply chain, a severe disaster in the Industrial city automatically triggered minor metric penalties in the neighboring cities, effectively illustrating the interconnected nature of global economics.

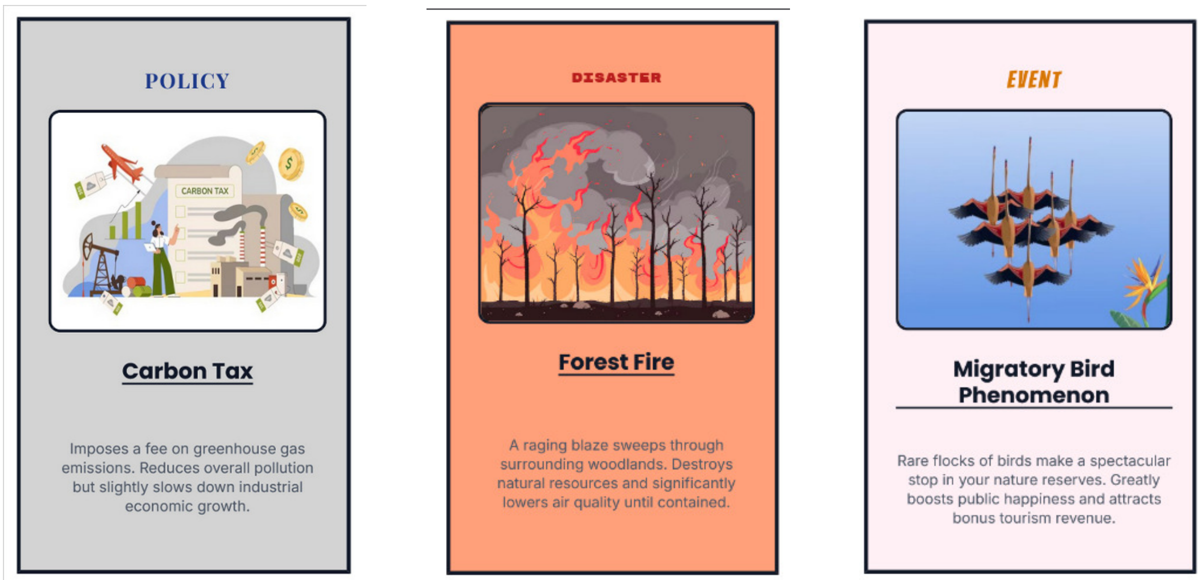

Figure 3: Policy, Disaster, Event Cards

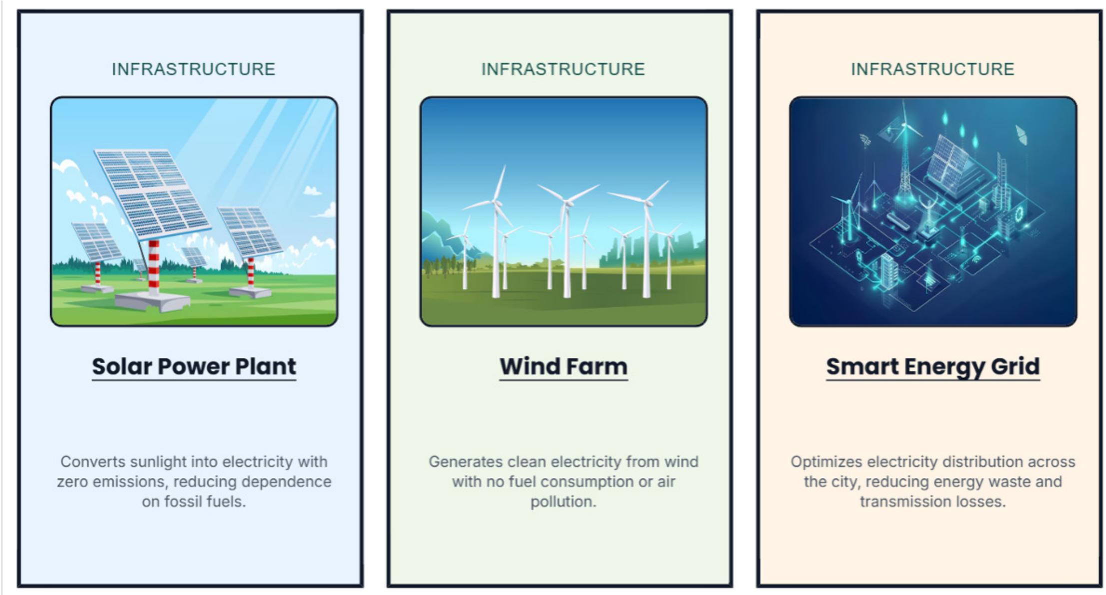

Figure 4: Infra Cards

Progression and the Two-Tiered Board

The physical board design—divided into an Outer Ring (Developing cities) and an Inner Ring (Developed cities)—functions effectively as a pacing mechanism. Cities begin with 1,000 points. The longer laps of the Outer Ring dictate that the effects of Policy and Disaster cards (which are tracked by lap count) endure for a longer real-time duration, simulating the protracted struggles of developing infrastructure.

Once a Mayor reaches the 4,000-point threshold, they ascend to the Inner Ring. The ESP32 seamlessly shifts their tracking logic to the shorter track geometry. The faster laps within the Inner Ring create a high-stakes endgame where economic income triggers frequently, but card effects expire rapidly. The game's win condition—completing exactly 6 laps within the Inner Ring—creates a tense race to optimize the weighted Total Score before the final lap concludes.

4 Conclusions

The Ecology Game successfully demonstrates that complex, multi-variable simulations regarding urban planning and environmental sustainability can be translated into an accessible, highly engaging tabletop format. By leveraging an ESP32 microcontroller, WebSockets, and a Flutter mobile application, the project circumvents the friction of manual bookkeeping that typically plagues intensive simulation board games.

The asymmetric faction weighting, dynamic disaster severity based on real-time sustainability metrics, and the cascading effects of global supply chain disruptions effectively mirror real-world governmental challenges. Participants are actively taught that raw economic output cannot guarantee sovereign success if livability and environmental health are neglected. Ultimately, this hybrid physical-digital architecture serves as a robust framework not only for this specific ecology simulation but as a pioneering model for the future of educational, "serious" tabletop gaming.